

## CS 4820/5820 – Homework 3 (Fall 2025)

### Logical Agents and Propositional Inference

Total: 12.5 pts (both levels)

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#### Submission Rules

- Use any language (Python/Java/C++/...). Implement inference yourself (no SAT/logic libraries).
- Submit one AAI-formatted PDF and your source code (with a README to run).
- Keep problems modest: small KBs, clear outputs. All figures must be original or recreated.
- Individual work. No late submissions.

#### Point Allocation (Undergraduate vs Graduate)

| Part                          | CS 4820 (UG)            | CS 5820 (Grad) |
|-------------------------------|-------------------------|----------------|
| A. KB Agents                  |                         |                |
| Propositional Basics          | 3.5                     | 3.0            |
| B. Inference Engine (Horn)    | 4.0                     | 4.0            |
| C. Wumpus Reasoning (2 moves) | 5.0                     | 3.5            |
| D. Graduate Extension         | <i>Bonus up to +2.0</i> | 2.0 (required) |
| <b>Total Counted</b>          | 12.5                    | 12.5           |

*Note:* Undergraduates may earn up to **+2.0 bonus** by attempting Part D; graduate students must complete Part D (still capped at 12.5 total).

### Part A: Knowledge-Based Agents & Propositional Logic (UG 3.5 pts, Grad 3.0 pts)

**Goal:** Practice small, precise propositional reasoning tasks.

a) **KB Agent Overview (0.5 pt)**

Briefly describe: KB, Tell/Ask, inference, percepts → actions.

b) **Equivalences (1.0 pt)**

For each pair, state whether logically equivalent. Justify with a 4-row truth table or laws.

- i)  $\neg(P \vee Q)$  vs.  $(\neg P) \wedge (\neg Q)$  (De Morgan)
- ii)  $(P \Rightarrow Q)$  vs.  $(\neg Q \Rightarrow \neg P)$  (Contraposition?)
- c) **Tiny Model Check (2.0/1.5 pts UG/Grad)**  
 KB:  $P \Rightarrow Q$ ,  $Q \Rightarrow R$ ,  $P$ . Using model checking on  $\{P, Q, R\}$ , decide if  $R$  is entailed. Show concise reasoning (no more than 6 lines).

## Part B: Inference Engine (Horn) & Tests (4.0 pts both levels)

**Goal:** Implement *one* Horn-clause inference method and test it on small KBs.

**B1. Implement One Algorithm (2.5 pts)** Choose and code exactly one:

- **Forward Chaining** (data-driven) on Horn rules.
- **Backward Chaining** (goal-driven) on Horn rules.

Input: KB as facts/rules (text or structured); Query: a ground atom. Output: entailment True/False and a brief trace (derived facts or goal stack expansion).

**B2. Test on Two KBs (1.5 pts)** Provide results for: (i) a tiny generic KB (3–5 rules), and (ii) a Wumpus fragment:

$$\neg B_{1,1}, \quad B_{2,1}, \quad B_{1,2}$$

$$B_{x,y} \Leftarrow P_{x+1,y} \vee P_{x-1,y} \vee P_{x,y+1} \vee P_{x,y-1}$$

Pose and answer queries like  $\neg P_{1,2}?$ ,  $\neg P_{2,1}?$  using your method. Keep domains to  $4 \times 4$  bounds.

## Part C: Wumpus Reasoning Agent (UG 5.0 pts, Grad 3.5 pts)

**Goal:** Use your engine to make *two* safe moves in a  $4 \times 4$  world.

**C1. Rules (0.5 pt)** Provide the exact Horn rules you use (you may simplify). For example:

$$\neg B(x, y) \Rightarrow (\neg P(x+1, y) \wedge \neg P(x-1, y) \wedge \neg P(x, y+1) \wedge \neg P(x, y-1)),$$

$$B(x, y) \Rightarrow P(x+1, y) \vee P(x-1, y) \vee P(x, y+1) \vee P(x, y-1).$$

(Clamp neighbors to the grid.) Define ‘ $OK(u, v) := (\neg P(u, v) \wedge \neg W(u, v))$ ’ if you also model stench.

**C2. Agent Loop (3.0/2.0 pts UG/Grad)** Start at (1,1). For exactly two steps:

1. Sense percepts at current cell: add  $B/\neg B$  (and  $S/\neg S$  if you include Wumpus).
2. Run your Part B engine to infer which neighbors are safe (entailed  $\neg P$  and  $\neg W$  if modeled).
3. Move to an *entailed safe* unvisited neighbor. If none are entailed safe, stop and report.

Produce a short log: (cell, percepts)  $\rightarrow$  new facts  $\rightarrow$  entailed safe neighbors  $\rightarrow$  chosen move.

**C3. Mini Table (1.5/1.0 pts UG/Grad)** Include a 4×4 table marking: Visited, Safe (entailed), Unknown. A simple ASCII/LaTeX grid is fine.

## Part D: Graduate Extension (Grad 2.0 pts required; UG bonus up to +2.0)

Choose one (clearly indicate which):

- a) **Resolution Entailment (2.0 pts)** Implement propositional *resolution* for CNF. Convert a small Wumpus KB to CNF and test entailment for two queries (one entailed, one not). Show a brief clause trace.
- b) **Scaling Study (2.0 pts)** Using your Part B engine, measure runtime vs. KB size by adding  $k = \{5, 10, 15\}$  extra neutral facts/rules. Plot runtime (or number of inferences) vs.  $k$  and discuss where it slows.

*UG bonus:* Completing either option can add up to +2.0 points, not exceeding 12.5 total.

## Appendix: Quick Hints

- Keep KBs tiny; prefer Horn clauses for Parts B–C.
- For model checking in Part A, enumerate only  $\{P, Q, R\}$ ; be concise.
- In Part C, two moves are enough—focus on correct entailment of safe neighbors.